Absolute concentration robustness with multistability in bimolecular networks: three species suffice, dimension two does not

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Abstract. Kaihnsa, Nguyen and Shiu proved that an at-most-bimolecular reaction network in which absolute concentration robustness (ACR) and nondegenerate multistationarity coexist has at least 3 species, 3 reactant complexes, 5 complexes and a two-dimensional stoichiometric subspace, all four bounds being attained. They conjectured that each of these four lower bounds becomes strictly larger when multistationarity is replaced by multistability. We show that the conjecture holds for three of the four quantities and fails for the fourth. The main result is that a two-dimensional bimolecular mass-action system with ACR has at most one exponentially stable positive steady state in each stoichiometric compatibility class. The proof combines a normal form of the vector field on the ACR hyperplane, a connectedness property of the positive steady-state set that holds precisely because products are at most bimolecular, and the global ACR condition. Consequently ACR with multistability needs dimension at least 3, at least 4 reactant complexes and, by a kinetic-subspace argument and the deficiency-one theorem, at least 6 complexes. On the other hand, three species suffice: the six-reaction network $\{Z \to Y + Z,\ 2Z \to Z,\ Y \to Y + Z,\ Y + Z \to 2X,\ X + Z \to 2Z,\ 2X \to Y\}$ has ACR in X for all rate constants and, for suitable ones, two exponentially stable positive steady states, certified in exact arithmetic. It has 3 species, 6 complexes and dimension 3, so these three minima are attained. The minimal number of reactant complexes lies between 4 and 6; with exactly 4 it is impossible for 4 species and for six of the nine reactant structures that can occur, and with 3 species at least 6 reactions are needed.

The question

Kaihnsa, Nguyen and Shiu [1] study the minimal size of reaction networks in which ACR and multistationarity coexist. Their main result (Theorem 1.2 there) states that an at-most-bimolecular network G admitting a vector of rate constants for which the mass-action system has ACR and is nondegenerately multistationary has at least 3 species, at least 3 reactant complexes, at least 5 complexes and dimension at least 2; the network $\{X_1 + X_2 \to 2X_3,\ X_3 \to X_1,\ 2X_3 \to 2X_2\}$ attains all four bounds. Section 6 of the paper ends with the following conjecture:

> *"In the present work, our interest in multistationarity comes from the fact that it is a necessary condition for multistability. Another interesting direction, therefore, is to investigate conditions for coexistence of ACR and multistability, rather than multistationarity. The "minimal" networks in the current work admit only two positive steady states and are not multistable. Hence, we conjecture that the lower bounds (on dimension and the numbers of species, reactant complexes, and all complexes) for the coexistence of ACR and multistability are strictly larger than the bounds proven here for ACR and multistationarity."*

Theorem 1 (main result). Let G be an at-most-bimolecular reaction network and suppose that for some vector of positive rate constants κ the mass-action system (G, κ) has ACR and is multistable. Then

(1) the stoichiometric subspace of G has dimension at least 3;

(2) G has at least 4 reactant complexes;

(3) G has at least 6 complexes.

On the other hand, there is a bimolecular network with 3 species, 6 complexes and dimension 3 that has ACR for every vector of rate constants and, for an open set of them, two exponentially stable positive steady states in one compatibility class (Theorem 21). Hence the conjecture holds for dimension, reactant complexes and complexes, and fails for the number of species. The bounds 3 (dimension), 3 (species) and 6 (complexes) are attained; the minimal number of reactant complexes lies between 4 and 6 (Section 5).

	ACR + nondegenerate multistationarity [1]	ACR + multistability lower bound	ACR + multistability best example
dimension	2	3	3 ($\mathcal{N}_1$)
species	3	3	3 ($\mathcal{N}_1$)
reactant complexes	3	4	6 ($\mathcal{N}_1$)
complexes	5	6	6 ($\mathcal{N}_1$)

Table 2 / Minimal sizes of at-most-bimolecular networks. All bounds of the first column are attained by the network $\{X_1 + X_2 \to 2X_3,\ X_3 \to X_1,\ 2X_3 \to 2X_2\}$ of [1].

Part 1 is the substance of the paper (Theorem 15); parts 2 and 3 follow from it together with results of [1] and a short argument on the kinetic subspace (Section 3).

1 Setting

A reaction network G on species $X_1, \dots, X_n$ is a finite set of reactions $y \to y'$ between complexes $y, y' \in \mathbb{Z}^n_{\geq 0}$, $y \neq y'$. It is *at most bimolecular* (bimolecular for short) if every complex, reactant or product, satisfies $|y| := y_1 + \cdots + y_n \leq 2$. Given positive rate constants $\kappa = (\kappa_{y \to y'})$, the mass-action system is

$$\dot{x} = f(x) := \sum_{y \to y'} \kappa_{y \to y'}\, x^y\, (y' - y) = \sum_{y} x^y\, v_y, \qquad v_y := \sum_{y' \,:\, y \to y'} \kappa_{y \to y'} (y' - y), \tag{3}$$

the second sum running over reactant complexes. The stoichiometric subspace is $S = \operatorname{span}\{y' - y\}$, $s = \dim S$ is the dimension of G, and the stoichiometric compatibility classes are the sets $(x^0 + S) \cap \mathbb{R}^n_{\geq 0}$. A positive steady state x^* is *nondegenerate* if the restriction $Df(x^*)|_S \colon S \to S$ is invertible, and *exponentially stable* if all eigenvalues of $Df(x^*)|_S$ have negative real part; the latter implies the former.

Definition 4. Following Definition 2.6 [1], (G, κ) has *ACR in* X_i if it has a positive steady state and x_i takes the same value $c > 0$ at all positive steady states (in all compatibility classes). We call (G, κ) *multistable* if some compatibility class contains at least two exponentially stable positive steady states. Both properties

refer to the same κ.

This is the usual notion of stability in the literature on small multistable networks §2.2 [2]: a steady state is stable if it is nondegenerate and the nonzero eigenvalues of the Jacobian have negative real parts, which is equivalent to our definition. As in Theorem 1.2 [1], the question is existential: a network qualifies if one vector κ gives both properties.

Exponentially stable steady states are nondegenerate, so a multistable system is nondegenerately multistationary and every lower bound of [1] applies to ACR with multistability.

Lemma 5 (coefficient cones)**.** Let G be bimolecular and let v_y be the coefficient vector of the monomial x^y in (3). Then $(v_y)_i \geq 0$ for every $i \notin \operatorname{supp}(y)$, and $\sum_i (v_y)_i \leq 0$ whenever $|y| = 2$.

Proof. Each $y' - y$ with $y \to y'$ has $(y' - y)_i = y'_i \geq 0$ for $i \notin \operatorname{supp}(y)$, and $\sum_i (y' - y)_i = |y'| - |y| \leq 0$ if $|y| = 2$, because $|y'| \leq 2$. Both properties pass to nonnegative combinations. □

The first property is the familiar characterization of mass-action vector fields Lemma 2.3 [1]; the second one is where bimolecularity of the *products* enters, and it is essential below (Remark 16).

2 Dimension two

Throughout this section G is bimolecular with $\dim S = 2$, κ is fixed, (G, κ) has ACR in X_1 with value $c > 0$, f is the vector field (3) and $H := \{x \in \mathbb{R}^n : x_1 = c\}$. All positive steady states lie in H. For $x \in H$ we write $x = (c, x')$ with $x' \in \mathbb{R}^{n-1}$.

Lemma 6. If (G, κ) has a nondegenerate positive steady state, then $e_1 \notin S^\perp$.

Proof. Let x^* be a nondegenerate positive steady state and let W be a matrix whose $n - 2$ rows form a basis of $S^\perp$. The map $F(x) = (f(x), Wx) \in S \times \mathbb{R}^{n-2}$ has invertible derivative at x^*: if $Df(x^*)u = 0$ and $Wu = 0$ then $u \in S$, hence $u = 0$. By the inverse function theorem, for every T near Wx^* there is a positive steady state $x(T)$ near x^* with $Wx(T) = T$. If $e_1 \in S^\perp$ we may take e_1^T as a row of W; the corresponding coordinate of T equals $x_1(T)$, which can then be made different from c, contradicting ACR. □

Under the hypothesis of the lemma, $S \cap e_1^\perp$ is one-dimensional; we fix a vector ℓ spanning it. Every compatibility class meets H in a segment parallel to ℓ.

Lemma 7 (the field on the ACR hyperplane)**.** Suppose some compatibility class contains two distinct nondegenerate positive steady states p and q. Then there are $\alpha \in S \setminus \{0\}$ and a polynomial h of degree at most 2 such that $f(x) = h(x)\,\alpha$ for all $x \in H$. Moreover $q = p + \tau\ell$ for some $\tau \neq 0$, and $h_L(t) := h(p + t\ell)$ has degree exactly 2 with simple zeros $t = 0$ and $t = \tau$.

Proof. Choose a basis σ_1, σ_2 of S and write $f = g_1\sigma_1 + g_2\sigma_2$ with polynomials g_1, g_2 of degree at most 2. Since $Df(p)$ maps onto S, the set $f^{-1}(0)$ is, near p, a smooth

submanifold of dimension $n-2$, and by ACR its positive points lie in H. Thus the restrictions $g_1|_H$ and $g_2|_H$, viewed as polynomials on $H \cong \mathbb{R}^{n-1}$, vanish on a smooth hypersurface M of H.

We claim that they have a nonconstant common factor (or one of them vanishes identically, in which case the lemma holds with α the other basis vector). By Baire's theorem, an irreducible factor F_1 of $g_1|_H$ vanishes on a nonempty open subset M_1 of M, and an irreducible factor F_2 of $g_2|_H$ vanishes on a nonempty open subset $M_2 \subseteq M_1$. If F_1 and F_2 were not proportional, they would be coprime, their common complex zero set would have dimension at most $n-3$, and so would its real points; but it contains the $(n-2)$-dimensional manifold M_2.

If the common factor F has degree 2, then $g_i|_H = c_i F$ with constants c_i, and $f|_H = F\,(c_1\sigma_1 + c_2\sigma_2)$. If F has degree 1, then $g_i|_H = F\,m_i$ with $\deg m_i \le 1$. If m_1, m_2 are proportional we conclude as before. Otherwise $\{x \in H : m_1 = m_2 = 0\}$ is empty or an affine subspace of dimension at most $n-3$. A nondegenerate positive steady state y with $F(y) \neq 0$ would have a neighbourhood in $f^{-1}(0)$, of dimension $n-2$, contained in that set; so every nondegenerate positive steady state satisfies $F = 0$. The steady states of the class of p lie on the segment $(p + \mathbb{R}\ell) \cap \mathbb{R}^n_{>0}$, on which F is affine; vanishing at $p \neq q$, it vanishes on the whole segment, so p is not isolated in its class, which contradicts nondegeneracy (nondegenerate steady states are isolated in their class by the inverse function theorem).

Finally, $p, q \in H$ lie in the same class, so $q - p \in S \cap e_1^{\perp} = \mathbb{R}\ell$. The polynomial h_L vanishes at 0 and τ; if its degree were at most one it would vanish identically and p would again not be isolated. Hence $\deg h_L = 2$ and both zeros are simple. Likewise $f|_H \not\equiv 0$ (otherwise p would not be isolated), so the vector α obtained above is nonzero. □

From now on we assume that a class contains two distinct nondegenerate positive steady states p, q, and we use α, h, ℓ, τ and h_L from Lemma 7.

Lemma 8. If $\alpha_1 = 0$, then $Df|_S$ has a positive real eigenvalue at p or at q.

Proof. Then $\alpha \in S \cap e_1^{\perp}$, so $\alpha = \gamma\ell$ with $\gamma \neq 0$. For $x_t := p + t\ell \in H$ we have $f(x_t) = \gamma h_L(t)\ell$, hence $Df(x_t)\ell = \gamma h_L'(t)\ell$: ℓ is an eigenvector with eigenvalue $\gamma h_L'(t)$. Since $h_L'(0)$ and $h_L'(\tau)$ are nonzero with opposite signs, this eigenvalue is positive at p or at q. □

Normal form. If $\alpha_1 \neq 0$ we normalize $\alpha_1 = 1$. Since $\{\alpha, \ell\}$ is a basis of S and $\ell_1 = 0$, we can write uniquely

$$f = \psi\,\alpha + \varphi\,\ell, \qquad \psi = f_1. \tag{9}$$

On H we have $f = h\alpha$, so φ vanishes on H and $\psi|_H = h$. As $\deg\varphi \le 2$, there is an affine function P with $\varphi = (x_1 - c)\,P$. The positive steady states are the positive solutions of $\psi = 0$ and $(x_1 - c)P = 0$.

Lemma 10 (determinant on the ACR hyperplane)**.** If $\alpha_1 = 1$, then $\det(Df(x)|_S) = -P(x)\,(\nabla\psi(x)\cdot\ell)$ for every $x \in H$.

Proof. On H, $Df = \alpha\,\nabla\psi^{\mathsf{T}} + P\,\ell\,e_1^{\mathsf{T}}$. Hence $Df\,\alpha = (\nabla\psi\cdot\alpha)\alpha + P\ell$ and $Df\,\ell = (\nabla\psi\cdot\ell)\alpha$. In the basis (α, ℓ) the matrix of $Df|_S$ is $\left(\begin{smallmatrix} \nabla\psi\cdot\alpha & \nabla\psi\cdot\ell \\ P & 0 \end{smallmatrix}\right)$, with determinant $-P\,(\nabla\psi\cdot\ell)$. □

Corollary 11. If $\alpha_1 = 1$ and p, q are both exponentially stable, then $P(p)\,P(q) < 0$.

Proof. A 2×2 matrix whose eigenvalues have negative real parts has positive determinant. Here $\nabla\psi(p)\cdot\ell = h_L'(0)$ and $\nabla\psi(q)\cdot\ell = h_L'(\tau)$ are nonzero with opposite signs, so Lemma 10 forces $P(p)$ and $P(q)$ to have opposite signs. □

The remaining steps show that P cannot change sign along the positive steady states. Write $\psi_H(x') := \psi(c, x')$ and

$$V^+ := \{x' \in \mathbb{R}^{n-1}_{>0} \colon \psi_H(x') = 0\},$$

so that the positive steady states are exactly the points (c, x') with $x' \in V^+$.

Lemma 12 (bimolecularity makes V^+ connected)**.** Assume $\alpha_1 = 1$. Then V^+ is connected, and $\nabla\psi_H \neq 0$ at every point of V^+.

Proof. For $2 \le j \le k \le n$ let a_{jk} be the coefficient of x_jx_k in $\psi = f_1$. Monomials containing x_1 become of degree at most one in x' on H, and $(x_1 - c)P$ contains no monomial x_jx_k with $j, k \ge 2$; hence the quadratic part of ψ_H is $Q(x') = \sum_{j\le k} a_{jk}x_jx_k$ and the coefficient vector of x_jx_k in f is $a_{jk}\alpha$. By Lemma 5 applied to $y = X_j + X_k$: $a_{jk} \ge 0$ (coordinate 1), and if $a_{jk} > 0$ then $\alpha_i \ge 0$ for all $i \notin \{1, j, k\}$ and $1+\sum_{i\ge2}\alpha_i \le 0$, so that $\sum_{i\in\{j,k\}}\alpha_i \le -1 - \sum_{i\notin\{1,j,k\}}\alpha_i \le -1$ (a single term when $j = k$). Let $N := \{i \ge 2 \colon \alpha_i < 0\}$. Every monomial x_jx_k with $a_{jk} > 0$ therefore satisfies $N \subseteq \{j, k\}$ and $N \cap \{j, k\} \neq \emptyset$. Since $h_L(t) = \psi_H(p' + t\ell')$ has degree 2, $Q(\ell') \neq 0$, so $Q \neq 0$ and $1 \le |N| \le 2$.

Second, for $i \in N$ the component f_i restricted to H equals $\alpha_i\psi_H$; its monomials not containing x_i have nonnegative coefficients (setting $x_1 = c > 0$ preserves this property of f_i), so the monomials of ψ_H not containing x_i have nonpositive coefficients. Two cases remain.

Case $N = \{a, b\}$. Then $Q = \beta x_ax_b$ with $\beta > 0$, every other monomial of ψ_H lacks x_a or x_b, and $\psi_H = \beta x_ax_b - G_0 - G_ax_a - G_bx_b - \sum_{k\notin\{a,b\}} G_kx_k$ with all $G \ge 0$. Writing $F := G_0 + G_ax_a + \sum_{k\notin\{a,b\}} G_kx_k$, the equation $\psi_H = 0$ reads $x_b(\beta x_a - G_b) = F$. On the open orthant either $F > 0$, and then V^+ is the graph $x_b = F/(\beta x_a - G_b)$ over the convex set $\{\beta x_a > G_b\} \cap \mathbb{R}^{n-2}_{>0}$, with $\partial\psi_H/\partial x_b = \beta x_a - G_b > 0$; or $F \equiv 0$, and then $V^+ = \{x_a = G_b/\beta\} \cap \mathbb{R}^{n-1}_{>0}$ is convex, with $\partial\psi_H/\partial x_a = \beta x_b > 0$.

Case $N = \{a\}$. Then $Q = x_a\sum_{k\ge2}\beta_kx_k$ with all $\beta_k \ge 0$, not all zero, and $\psi_H = \beta_ax_a^2 + x_a(K + R(x'')) - G(x'')$, where x'' collects the coordinates other than x_1, x_a, $R(x'') = \sum_{k\neq a}\beta_kx_k$, $K \in \mathbb{R}$, and G is affine with nonnegative coefficients. On the open orthant either $G > 0$ or $G \equiv 0$. If $G > 0$ and $\beta_a > 0$, the quadratic in x_a has exactly one positive root, which is simple; if $G > 0$ and $\beta_a = 0$, the positive root $x_a = G/(K + R)$ exists exactly on the convex set $\{K + R > 0\}$, where $\partial\psi_H/\partial x_a = K + R > 0$. In both cases V^+ is the graph of a smooth function of x'' over a convex set. If $G \equiv 0$, then $V^+ = \{\beta_ax_a+K+R(x'') = 0\}\cap\mathbb{R}^{n-1}_{>0}$ is convex, and on it $\nabla\psi_H = x_a\nabla(\beta_ax_a+K+R) \neq 0$, because β_a or some β_k with $k \neq a$ is positive. □

Lemma 13 (global ACR)**.** Assume $\alpha_1 = 1$ and $P(p)P(q) < 0$, and write $P_H(x') := P(c, x')$. Then $Z := \{x' \in \mathbb{R}^{n-1}_{>0} \colon P_H(x') = 0\}$ is nonempty, and there is $\varepsilon \in \{1, -1\}$ with $\varepsilon\,\psi_H \ge 0$ on Z.

Proof. P_H takes both signs on the convex set $\mathbb{R}^{n-1}_{>0}$, so $Z \neq \emptyset$; in particular P depends on some x_k with $k \geq 2$, and the hyperplane $\Pi := \{P = 0\}$ of $\mathbb{R}^n$ is transverse to H. At a positive point of Π we have $f = \psi\alpha$, so by ACR ψ has no zero on the convex sets $\Pi^{\pm} := \Pi \cap \mathbb{R}^n_{>0} \cap \{\pm(x_1 - c) > 0\}$, and hence has a constant sign $\varepsilon_{\pm}$ on each. Both sets accumulate at every point of $\{c\} \times Z$ (move inside Π along $e_1 - (\pi_1/\pi_k)e_k$, where π_j are the coefficients of P and $\pi_k \neq 0$), so $\varepsilon_+\psi_H \geq 0$ and $\varepsilon_-\psi_H \geq 0$ on Z. If $\varepsilon_+ \neq \varepsilon_-$, then ψ_H vanishes on Z, a nonempty open subset of the hyperplane $\{P_H = 0\}$, so P_H divides ψ_H: $\psi_H = P_H\, A$ with A affine. Since $P_H(p'), P_H(q') \neq 0$, A vanishes at p' and q', hence on the line through them, and $h_L \equiv 0$, contradicting Lemma 7. So $\varepsilon := \varepsilon_+ = \varepsilon_-$ works. □

Lemma 14 (no transversal sign change)**.** In the situation of Lemma 13, let $z \in V^+ \cap Z$. Then $\nabla\psi_H(z) = \mu\,\nabla P_H$ for some $\mu \neq 0$, and z has a neighbourhood U in V^+ on which $-\operatorname{sign}(\mu)\,\varepsilon\,P_H \geq 0$.

Proof. Let $E := \{P_H = 0\}$, with direction space T. The function $\varepsilon\psi_H$ is nonnegative on the relatively open set $Z \subseteq E$ and vanishes at z, so $\nabla\psi_H(z) \perp T$, that is $\nabla\psi_H(z) = \mu\nabla P_H$; and $\mu \neq 0$ by Lemma 12. Let Q be the quadratic part of ψ_H and B its polar form. For $v \in T$, $\psi_H(z+v) = Q(v)$, so $\varepsilon Q \geq 0$ on T. Fix ν with $\nabla P_H \cdot \nu = 1$ and write $w = v + s\nu$ with $v \in T$, $s = P_H(z+w)$. If $z + w \in V^+$,

$$0 = \psi_H(z+w) = \mu s + Q(v) + 2sB(v,\nu) + s^2Q(\nu), \quad \text{i.e.} \quad s\,(\mu + 2B(v,\nu) + sQ(\nu)) = -Q(v).$$

For w small the bracket has the sign of μ, so s is zero or has the sign of $-\mu\varepsilon$. □

Theorem 15 (dimension two)**.** Let G be an at-most-bimolecular network of dimension 2 and let κ be such that (G, κ) has ACR. Then every stoichiometric compatibility class contains at most one exponentially stable positive steady state. In particular (G, κ) is not multistable.

Proof. Suppose $p \neq q$ are exponentially stable positive steady states in one class; they are nondegenerate. By Lemmas 6 and 7 we have the objects α, ℓ, h; by Lemma 8, $\alpha_1 \neq 0$, and we normalize $\alpha_1 = 1$. By Corollary 11, $P(p)P(q) < 0$. Define $\theta\colon V^+ \to \{1, -1\}$ by $\theta(x') = \operatorname{sign} P_H(x')$ if $P_H(x') \neq 0$ and $\theta(x') = -\operatorname{sign}(\mu(x'))\,\varepsilon$ if $x' \in Z$, where $\mu(x') = \nabla\psi_H(x') \cdot \nabla P_H / |\nabla P_H|^2$ is continuous and nonzero on $V^+ \cap Z$ (Lemma 14). Then θ is locally constant: near a point with $P_H \neq 0$ by continuity, and near $z \in V^+ \cap Z$ by Lemma 14 applied on a neighbourhood where μ keeps its sign. As V^+ is connected (Lemma 12), θ is constant, so $P(p)$ and $P(q)$ have the same sign, a contradiction. □

Remark 16 (where the bimolecular products matter)**.** The bound $|y'| \leq 2$ on product complexes is used only through the second property in Lemma 5, inside Lemma 12. Without it, ψ_H may contain the terms $x_2^2 - 3x_2 + 2$, and V^+ may consist of two parallel sheets on which P has opposite signs; the argument then breaks down. In the planar model $\dot{u} = t^2 - 1 - u/c$, $\dot{t} = -ut$ (with $u = x_1 - c$), the two stable nodes $t = \pm 1$ of the ACR line $u = 0$ are separated by the invariant line $t = 0$, which ends at a saddle on the face $x_1 = 0$. We have not found a mass-action realization of this mechanism with bimolecular reactant complexes (linear programs with the constraints of Lemma 5 minus the product bound are feasible locally, but the global ACR condition failed on every sample), so we do not know whether Theorem 15 extends to networks with larger

product complexes.

3 Reactant complexes and complexes

Corollary 17 (dimension and reactant complexes)**.** If a bimolecular network G admits κ with ACR and multistability, then $\dim S \geq 3$ and G has at least 4 reactant complexes.

Proof. One-dimensional networks never have ACR with nondegenerate multistationarity Proposition 3.10 [1], and dimension 2 is excluded by Theorem 15. A bimolecular network with ACR and nondegenerate multistationarity has at least 3 species Theorem 4.1 [1]. If G has $k \geq 1$ independent conservation laws, it has at least $n - k + 1 = \dim S + 1 \geq 4$ reactant complexes Theorem 5.5 [1]; if it is full-dimensional, at least $n+2 \geq 5$ Theorem 5.1 [1]. □

For the number of complexes we use the classical objects of chemical reaction network theory: m complexes, ℓ linkage classes, t terminal strong linkage classes, deficiency $\delta = m - \ell - s$, and the factorization $f(x) = Y A_\kappa \Psi(x)$, where Y is the $n \times m$ matrix of complexes, $\Psi(x) = (x^y)_y$ and A_κ is the Laplacian matrix of the weighted reaction graph [3]. The kinetic subspace $K_\kappa := Y(\operatorname{Im} A_\kappa) \subseteq S$ contains every value of f and every column of Df.

Lemma 18 (kinetic subspace)**.** If (G, κ) has a nondegenerate positive steady state and G has a nonterminal complex, then $\delta \geq t - \ell + 1$.

Proof. At a nondegenerate positive steady state x^* the image of $Df(x^*)|_S$ is S, so $K_\kappa = S$. The kernel of A_κ has dimension t and consists of vectors supported on terminal complexes, so $\dim \operatorname{Im} A_\kappa = m - t$, and $A_\kappa \Psi(x^*) \neq 0$ because $\Psi(x^*)$ is positive on a nonterminal complex. As $Y A_\kappa \Psi(x^*) = f(x^*) = 0$, the map Y has a nonzero kernel on $\operatorname{Im} A_\kappa$. Hence $s = \dim K_\kappa \leq m - t - 1$, that is $\delta = m - \ell - s \geq t - \ell + 1$. □

Corollary 19 (complexes)**.** If a bimolecular network G admits κ with ACR and multistability, then G has at least 6 complexes.

Proof. By Corollary 17, $s \geq 3$. A network of deficiency zero is not multistationary Lemma 2.4 [1], so $\delta \geq 1$ and $m = \ell + s + \delta \geq 5$, with equality only if $\ell = 1$, $s = 3$, $\delta = 1$. Suppose $m = 5$. If G has a nonterminal complex, Lemma 18 gives $t \leq \ell + \delta - 1 = 1$; otherwise all complexes are terminal and $t = \ell = 1$. So G has one linkage class, of deficiency one, with a single terminal strong linkage class, and the deficiency-one theorem [4] gives at most one positive steady state in each compatibility class. This contradicts multistability. □

Remark 20 (on the deficiency-one theorem)**.** The uniqueness part of the deficiency-one theorem holds without weak reversibility, provided each linkage class contains exactly one terminal strong linkage class, $\delta_\theta \leq 1$ for each linkage class and $\sum_\theta \delta_\theta = \delta$ [4]. Kaihnsa, Nguyen and Shiu [Lemma 2.5 [1]] state it for weakly reversible networks only, but their proof of the bound of 5 complexes (Theorem 4.9(2) there) applies it to networks that need not be weakly reversible; Lemma 18 supplies the missing hypothesis $t = 1$ there, so their bound stands. The same argument shows that two nondegenerate positive steady states in one class force $m \geq s + 3$ for networks of any molecularity: $\delta \geq 1$ always,

and $\delta \geq 2$ when $\ell = 1$.

4 Three species and six complexes suffice

Theorem 21 (three species, six complexes)**.** The bimolecular network on species X, Y, Z

$$\mathcal{N}_1: \quad Z \xrightarrow{k_1} Y+Z, \quad 2Z \xrightarrow{k_2} Z, \quad Y \xrightarrow{k_3} Y+Z, \quad Y+Z \xrightarrow{k_4} 2X, \quad X+Z \xrightarrow{k_5} 2Z, \quad 2X \xrightarrow{k_6} Y$$

has 3 species, 6 reactions, 6 reactant complexes, 6 complexes and dimension 3. For every vector of rate constants it has ACR in X, with value $2k_1/k_5$. For $(k_1, \ldots, k_6) = (5, 1, 1, 4, 10, 7)$ its positive steady states are exactly

$$P_1 = (1, \tfrac{19}{4}, \tfrac{1}{2}), \qquad P_2 = (1, 3, 1), \qquad P_3 = (1, \tfrac{7}{4}, \tfrac{7}{2}),$$

and P_1, P_3 are exponentially stable while P_2 is a saddle. The network being full-dimensional, all three lie in the unique compatibility class $\mathbb{R}^3_{\geq 0}$. Multistability and ACR persist for all rate constants in a neighbourhood of this vector.

Proof. The mass-action system is

$$\begin{aligned} \dot{x} &= 2k_4yz - k_5xz - 2k_6x^2, \\ \dot{y} &= k_1z - k_4yz + k_6x^2, \\ \dot{z} &= -k_2z^2 + k_3y - k_4yz + k_5xz. \end{aligned}$$

The reaction vectors $(0,1,0)$, $(0,0,-1)$, $(2,-1,-1)$ and $(-1,0,1)$ span $\mathbb{R}^3$. The weight $w = (1,2,0)$ gives $\dot{x} + 2\dot{y} = z\,(2k_1 - k_5x)$, so every positive steady state has $x = c := 2k_1/k_5$. On this plane $\dot{y} = 0$ gives $y = (k_1z + k_6c^2)/(k_4z)$, and then

$$k_4z\,\dot{z} = -k_2k_4z^3 + k_1k_4z^2 + (k_1k_3 - k_4k_6c^2)\,z + k_3k_6c^2,$$

which is positive at $z = 0$ and negative for large z. So a positive steady state exists for every κ, and ACR holds for every κ. For the given rate constants $c = 1$, $y = (5z+7)/(4z)$ and $4z\,\dot{z} = -(2z-1)(z-1)(2z-7)$, which gives exactly P_1, P_2, P_3. The Jacobian matrices and their characteristic polynomials are

$$\begin{aligned} J(P_1) &= \begin{pmatrix} -33 & 4 & 28 \\ 14 & -2 & -14 \\ 5 & -1 & -10 \end{pmatrix}, & \chi_1 &= \lambda^3 + 45\lambda^2 + 206\lambda + 30 = (\lambda+5)(\lambda^2 + 40\lambda + 6), \\ J(P_2) &= \begin{pmatrix} -38 & 8 & 14 \\ 14 & -4 & -7 \\ 10 & -3 & -4 \end{pmatrix}, & \chi_2 &= \lambda^3 + 46\lambda^2 + 47\lambda - 50, \\ J(P_3) &= \begin{pmatrix} -63 & 28 & 4 \\ 14 & -14 & -2 \\ 35 & -13 & -4 \end{pmatrix}, & \chi_3 &= \lambda^3 + 81\lambda^2 + 632\lambda + 1050. \end{aligned}$$

By the Routh–Hurwitz criterion ($45 \cdot 206 > 30$ and $81 \cdot 632 > 1050$, all coefficients positive) P_1 and P_3 are exponentially stable; $\chi_2(0) = -50 < 0$ gives P_2 a positive real eigenvalue. All three steady states are nondegenerate, so they persist, with their stability types, for nearby rate constants, and ACR holds for all rate constants. □

The mechanism is a weighted integral feedback: the pair of complexes $\{Z,\ X+Z\}$ is the only one whose reactions change $w \cdot x = x + 2y$, so that $w \cdot f = z\,(2k_1 - k_5x)$. Two further certified networks with six complexes, one of them weakly reversible with two

linkage classes of deficiency zero and total deficiency one, are given in the sub-study `complexes/report.md`.

5 Further results and open questions

The following results complete Table 2. Their proofs, partly computer-assisted with exact arithmetic, are in the two sub-study reports `examples/report.md` and `complexes/report.md`, with the scripts that check them.

(1) *A second three-species example with rational steady states.* The network $\mathcal{G}_7 = \{X+Y \xrightarrow{1} 2Y,\ Y \xrightarrow{1} 2Z,\ 0 \xrightarrow{5} X+Z,\ X+Z \xrightarrow{4} 2Z,\ 2Z \xrightarrow{1} X+Z,\ Y+Z \xrightarrow{6} Y\}$ has $\dot{y} = y(k_1 x - k_2)$, hence ACR in X for all rate constants, and for the indicated ones exactly three positive steady states $(1,2,1)$, $(1, \frac{5}{4}, \frac{3}{2})$, $(1, \frac{10}{9}, \frac{5}{3})$, the first and the last exponentially stable (exact Gröbner basis and Routh–Hurwitz). It has 7 complexes.

(2) *Six complexes in two linkage classes.* The weakly reversible network $\{2Z \rightleftarrows X+Z,\ Y+Z \to X,\ Y+Z \to X+Y,\ X \to 2X,\ X+Y \to X,\ 2X \to Y+Z\}$ has 6 complexes, two linkage classes of deficiency 0 and total deficiency 1, ACR in X and, for suitable rate constants, two exponentially stable positive steady states. So the deficiency-one theorem cannot be applied to two linkage classes when $\sum_\theta \delta_\theta < \delta$, and six-complex examples need not have a single linkage class.

(3) *Reactions with three species.* No full-dimensional bimolecular network with 3 species and at most 5 reactions has three nondegenerate positive steady states, or two exponentially stable ones, for any rate constants. Networks with at most 4 distinct reactant complexes are covered by Propositions 3.1(3) and 3.5 [1]; the 3 188 646 networks with 5 distinct reactant complexes were enumerated exactly, through a reduction to the sign changes of one univariate function whose derivative has, up to a fixed sign, the sign of the Jacobian determinant (two exponentially stable states have determinants of the same sign, which forces a third steady state between them). By Corollary 17, a three-species network with ACR and multistability is full-dimensional, so it needs at least 6 reactions, which $\mathcal{N}_1$ and $\mathcal{G}_7$ attain. The restriction to full-dimensional networks is necessary: the two-dimensional network $\{0 \xrightarrow{10} Z,\ Y+Z \xrightarrow{10} X,\ X+Z \xrightarrow{1} Y+Z,\ X+Y \xrightarrow{1} 2Y\}$ has the three nondegenerate positive steady states $(5,1,1)$, $(4,2,\frac{1}{2})$, $(3,3,\frac{1}{3})$ in the class $x+y=6$, the outer two exponentially stable (it has no ACR).

(4) *Four reactant complexes.* With exactly 4 reactant complexes and ACR, the positive steady states form a coset of a toric variety, and the number of them in a class is bounded by a rule of signs for exponential sums. For 4 species no class contains two exponentially stable positive steady states (two such states would force a third one, and the rule of signs allows at most two). For 5 or more species exactly nine reactant structures are compatible with ACR (none with 7 or 8 species); six of them are excluded by proof, and the remaining three were excluded only numerically, by sampling all realizable stoichiometric subspaces.

Open questions. Is the minimal number of reactant complexes 4, 5 or 6? For full-dimensional networks on three species, is the bound $n+2=5$ of Theorem 5.1 [1] attained with multistability? Does Theorem 15 hold for networks with bimolecular reactant

complexes but larger products (Remark 16)? Finally, [5] found ACR with bistability in a dual-site futile cycle with bifunctional enzymes (six species, dimension four); our examples show that much smaller mechanisms, based on a weighted integral feedback, suffice.

6 Computer verification

All scripts print `PASS`/`FAIL` for each claim and exit with a nonzero code on failure. They were run under WSL (Fedora 44) with Python 3.12, sympy 1.14 and python-flint 0.9.

- `scripts/check_dimension_two_example.py` (8 s): on 20 random rational instances of the KNS network of Example 4.10, checks the determinant formula of Lemma 10 and that the two positive steady states of a class have determinants of opposite signs. This is a sanity check; Theorem 15 is proved by hand.
- `complexes/scripts/certify_examples.py` (exact rational arithmetic, standard library only, under 0.1 s): Theorem 21 and the two further six-complex examples (ACR identity, steady states, characteristic polynomials, Routh–Hurwitz conditions, counts of complexes and deficiency). `complexes/scripts/eigen_enclosures.py`: eigenvalue enclosures in ball arithmetic.
- `examples/scripts/certify_example.py` (under 1 s): the network $\mathcal{G}_7$ (exact Gröbner basis, root isolation, ball-arithmetic eigenvalues). `examples/scripts/enum5_reactions.py` (about two minutes on 12 cores): the enumeration of item 3. `examples/scripts/e2_reactant_sets.py` and `check_gale_formula.py`: the reactant structures of item 4 and the determinant formula used there.

Searches and simulations used to find the examples are in `explore` sub-folders; they are floating-point and not part of any proof.

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